

Proiect Retele de Calculatoare

Proiectare si implementare retea LAN – SmartOffice

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1 Introducere

Acest proiect prezinta realizarea unei retele locale (LAN) pentru o companie de tip SmartOffice. Reteaua a fost proiectata astfel incat sa fie bine organizata, usor de administrat si sigura, folosind concepte studiate la disciplina Retele de Calculatoare.

Scopul principal este separarea traficului pe departamente, asigurarea comunicarii intre acestea si oferirea accesului controlat catre o retea externa.

2 Scenariul retelei

Compania este organizata pe mai multe departamente:

- Administrativ (ADMIN)
- Financiar si Resurse Umane (FIN_HR)
- Vanzari (SALES)
- IT
- Telefonie IP (VOIP)
- Retea wireless pentru vizitatori (GUEST_WIFI)

Pentru fiecare departament a fost creat un VLAN separat, pentru o mai buna organizare si cresterea securitatii retelei.

3 Proiectarea retelei

3.1 Topologia retelei

Topologia utilizata este de tip star, avand un router central conectat la mai multe switch-uri. Aceasta solutie permite o administrare simpla si o extindere usoara a retelei.

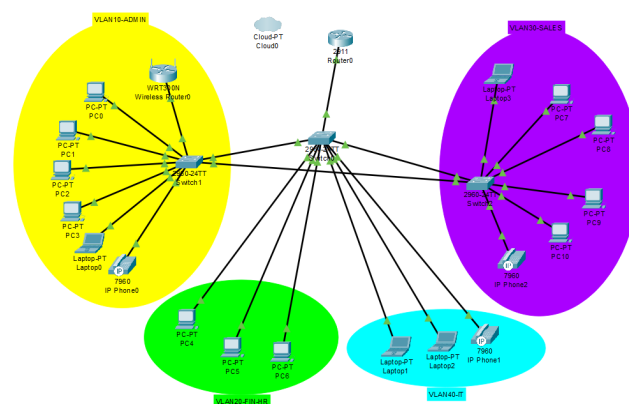


Figura 1: Topologia retelei realizata in Cisco Packet Tracer

3.2 Echipamente utilizate

- Router Cisco 2911
- Switch-uri Cisco 2960
- PC-uri si laptopuri
- Telefoane IP
- Router wireless / Access Point
- Cloud (simulare Internet)

3.3 Inventarul echipamentelor

VLAN	Departament	Echipamente
10	ADMIN	4 PC-uri, 1 laptop, 1 telefon IP
20	FIN_HR	3 PC-uri
30	SALES	4 PC-uri, 1 laptop, 1 telefon IP
40	IT	2 laptopuri, 1 telefon IP
50	VOIP	Telefoane IP
99	GUEST_WIFI	Router wireless / AP

4 Plan de adresare IP

4.1 Subnetare IPv4

VLAN	Retea	Masca	Gateway	Broadcast
10	192.168.10.0	/24	192.168.10.1	192.168.10.255
20	192.168.20.0	/24	192.168.20.1	192.168.20.255
30	192.168.30.0	/24	192.168.30.1	192.168.30.255
40	192.168.40.0	/24	192.168.40.1	192.168.40.255
50	192.168.50.0	/24	192.168.50.1	192.168.50.255
99	192.168.99.0	/24	192.168.99.1	192.168.99.255

5 Configurarea echipamentelor

5.1 VLAN-uri si trunk-uri

```
Switch0>show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Gig0/2
10 ADMIN	active	
20 FIN_HR	active	Fa0/1, Fa0/2, Fa0/3
30 SALES	active	
40 IT	active	Fa0/4, Fa0/5, Fa0/7
50 VOIP	active	Fa0/6, Fa0/8
99 GUEST_WIFI	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch0>show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/23	on	802.1q	trunking	1
Fa0/24	on	802.1q	trunking	1
Gig0/1	on	802.1q	trunking	1

```
Port Vlan allowed on trunk
```

Fa0/23	10,20,30,40,50,99
Fa0/24	10,20,30,40,50,99
Gig0/1	10,20,30,40,50,99

```
Port Vlan allowed and active in management domain
```

Fa0/23	10,20,30,40,50,99
Fa0/24	10,20,30,40,50,99
Gig0/1	10,20,30,40,50,99

```
Port Vlan in spanning tree forwarding state and not pruned
```

Fa0/23	10,20,30,40,50,99
Fa0/24	10,20,30,40,50,99
Gig0/1	10,20,30,40,50,99

```
Switch0>  
Switch0>
```

Figura 2: Configurarea VLAN-urilor si a trunk-urilor

5.2 Rutare inter-VLAN

```
Router0>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L    192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L    192.168.50.1/32 is directly connected, GigabitEthernet0/0.50
192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.99.0/24 is directly connected, GigabitEthernet0/0.99
L    192.168.99.1/32 is directly connected, GigabitEthernet0/0.99
```

Figura 3: Tabela de rutare pe router

5.3 DHCP

```
Router0>show ip dhcp binding
IP address      Client-ID/
                Hardware address
192.168.10.15    0010.1165.DC77    --    Automatic
192.168.10.12    0001.C93B.A4AE    --    Automatic
192.168.10.13    00E0.A3A7.C752    --    Automatic
192.168.10.17    000C.CFC4.01E7    --    Automatic
192.168.10.21    0001.C770.253B    --    Automatic
192.168.20.12    00D0.58BB.729A    --    Automatic
192.168.20.13    00D0.D36B.73E8    --    Automatic
192.168.20.11    0001.42C8.E5A7    --    Automatic
192.168.30.11    0090.2B55.ECB6    --    Automatic
192.168.30.12    00E0.F79B.8D88    --    Automatic
192.168.30.14    000C.CFDE.9C57    --    Automatic
192.168.30.13    0090.2BD7.8270    --    Automatic
192.168.30.15    0004.9A72.2645    --    Automatic
192.168.40.11    00D0.9702.C32D    --    Automatic
192.168.40.12    00E0.B06A.3443    --    Automatic
192.168.99.16    0000.0C12.0601    --    Automatic
Router0>
```

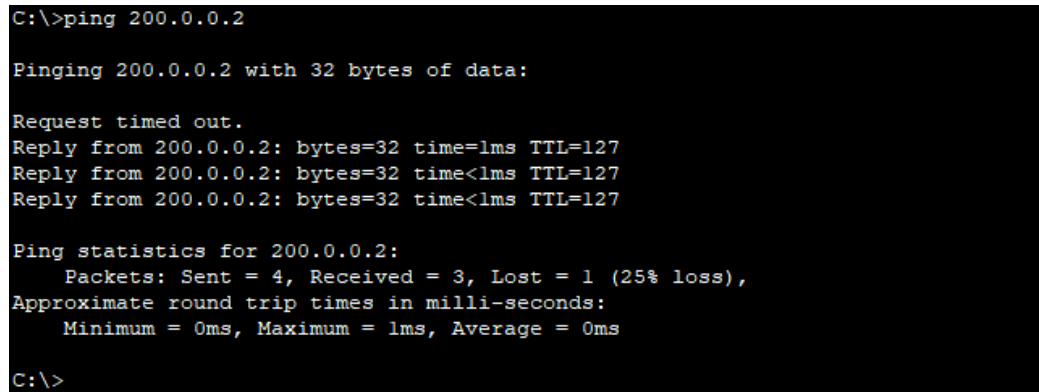
Figura 4: Alocarea adreselor IP prin DHCP

5.4 NAT

```
Router0>show ip nat translations
Router0>
Router0>
Router0>
Router0>
Router0>
Router0>
Router0>show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.0.0.1:17       192.168.10.22:17  200.0.0.2:17       200.0.0.2:17
icmp 200.0.0.1:18       192.168.10.22:18  200.0.0.2:18       200.0.0.2:18
icmp 200.0.0.1:19       192.168.10.22:19  200.0.0.2:19       200.0.0.2:19
icmp 200.0.0.1:20       192.168.10.22:20  200.0.0.2:20       200.0.0.2:20
```

Figura 5: Verificarea translatiilor NAT

6 Testarea retelei



```
C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 200.0.0.2: bytes=32 time=1ms TTL=127
Reply from 200.0.0.2: bytes=32 time<1ms TTL=127
Reply from 200.0.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figura 6: Test de conectivitate intre retele

7 Servicii suplimentare si creativitate

7.1 Spanning Tree Protocol

```
Switch0>show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0090.0C53.6BC2
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4097 (priority 4096 sys-id-ext 1)
             Address     0090.0C53.6BC2
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/23                   Desg FWD 19      128.23   P2p
Fa0/24                   Desg FWD 19      128.24   P2p
Gi0/1                    Desg FWD 4       128.25   P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    10
             Address     00D0.97BB.AA00
             Cost         19
             Port         24(FastEthernet0/24)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4106 (priority 4096 sys-id-ext 10)
             Address     0090.0C53.6BC2
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/23                   Desg FWD 19      128.23   P2p
Fa0/24                   Root FWD 19      128.24   P2p
Gi0/1                    Desg FWD 4       128.25   P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    4116
             Address     0090.0C53.6BC2
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4116 (priority 4096 sys-id-ext 20)
             Address     0090.0C53.6BC2
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                   Desg FWD 19      128.1    P2p
Fa0/2                   Desg FWD 19      128.2    P2p
Fa0/3                   Desg FWD 19      128.3    P2p
Fa0/23                  Desg FWD 19      128.23   P2p
Fa0/24                  Desg FWD 19      128.24   P2p
```

Figura 7: Verificarea STP

7.2 VLAN Trunking Protocol

```
Switch0>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : proiectLAN
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : 00D0.BA70.8E00
Configuration last modified by 0.0.0.0 at 3-1-93 03:43:53
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 255
Number of existing VLANs : 11
Configuration Revision    : 15
MD5 digest                : 0xEA 0xF0 0x2F 0xD5 0x65 0xB4 0x5E 0x8E
                          : 0xB3 0xCB 0x4D 0x65 0x04 0x7B 0x8D 0x19
```

Figura 8: Status VTP

7.3 Control ierarhic al accesului

Pentru cresterea securitatii, accesul la router a fost configurat pe baza de utilizatori, folosind autentificare VTY (Telnet). Au fost definite trei niveluri de acces.

Utilizator	Privilege Level	Rol
admin	15	Administrare completa
operator	5	Monitorizare si comenzi de baza
guest	1	Acces limitat (vizualizare)

```
Router0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#
Router0(config)#enable secret admin123
Router0(config)#
Router0(config)#username admin privilege 15 secret admin123
Router0(config)#username operator privilege 5 secret operator123
Router0(config)#username guest privilege 1 secret guest123
Router0(config)#
Router0(config)#line vty 0 4
Router0(config-line)# login local
Router0(config-line)# transport input telnet
Router0(config-line)#end
Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#no enable secret
Router0(config)#end
Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#username admin privilege 15 secret admin123
Router0(config)#username operator privilege 5 secret operator123
Router0(config)#username guest privilege 1 secret guest123
Router0(config)#end
Router0#
%SYS-5-CONFIG_I: Configured from console by console
```

Figura 9: Configurarea nivelurilor de acces pe router

```

Username: guest
Password:
Router0>telnet 192.168.10.1
Trying 192.168.10.1 ...Open

User Access Verification

Username: admin
Password:
Router0#telnet 192.168.10.1
Trying 192.168.10.1 ...Open

User Access Verification

Username: operator
Password:
% Login invalid

Username: operator
Password:
Router0#

```

Figura 10: Autentificare Telnet folosind utilizatori diferiti

8 Comenzi utilizate pentru verificare

Functionalitate	Comanda
VLAN-uri	show vlan brief
Trunk-uri	show interfaces trunk
Rutare	show ip route
DHCP	show ip dhcp binding
NAT	show ip nat translations
STP	show spanning-tree
VTP	show vtp status
Acces VTY	telnet / login local

9 Concluzii

Proiectul realizat indeplineste cerintele impuse. Reteaua este functionala, bine structurata si securizata, folosind VLAN-uri, rutare inter-VLAN, DHCP, NAT si control ierarhic al accesului.